

Question Number	Answer	Additional Guidance	Mark
3a	Answer that includes the following: there is no (significant) difference between the (mean) number of mayfly nymphs in stream A and stream B	Accept in polluted and unpolluted streams	Exp 1

Question Number	Answer	Additional Guidance	Mark																																						
3b	An answer that includes the following: - suitable table format with data (1) - correct column headings (1) - means correctly calculated ie 36.0 and 31.6 or 36 and 32 (1)	Example table <table><tr><td rowspan="17"></td><td colspan="2">Number of mayfly nymphs</td></tr><tr><td>stream A</td><td>stream B</td></tr><tr><td>27</td><td>25</td></tr><tr><td>37</td><td>16</td></tr><tr><td>24</td><td>34</td></tr><tr><td>45</td><td>12</td></tr><tr><td>34</td><td>35</td></tr><tr><td>38</td><td>26</td></tr><tr><td>49</td><td>43</td></tr><tr><td>61</td><td>51</td></tr><tr><td>40</td><td>26</td></tr><tr><td>20</td><td>24</td></tr><tr><td>28</td><td>36</td></tr><tr><td>38</td><td>26</td></tr><tr><td>42</td><td>49</td></tr><tr><td>25</td><td>52</td></tr><tr><td>32</td><td>19</td></tr><tr><td>mean</td><td>36.0 / 36</td><td>31.6 / 32</td></tr></table>		Number of mayfly nymphs		stream A	stream B	27	25	37	16	24	34	45	12	34	35	38	26	49	43	61	51	40	26	20	24	28	36	38	26	42	49	25	52	32	19	mean	36.0 / 36	31.6 / 32	Exp 3
	Number of mayfly nymphs																																								
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3c	An answer that includes the following: - bar graph with linear scale and axes labelled with units (1) - means plotted correctly (1) - range bars plotted correctly (1)	Must start at zero ie not broken axis Mean number of mayfly nymphs and (stream) A and B Accept ECF from 3b	Ex

Question Number	Answer	Additional Guidance	Mark
3di	An answer that includes the following: - correct calculation of numerator (1) - correct substitution of given $(S_A)^2$ and $(S_B)^2$ (1) - correct value of t (1)	Example calculation: $4.4 / 36.0 - 31.6$ or $4.0 / 36 - 32$ Ecf if wrong means used $\frac{116}{15} + \frac{160}{15}$ $t = 1.026 / 1.03$ or $0.933 / 0.93$ Correct answer with no working gains full marks	Exp 3

Question Number	Answer	Additional Guidance	Mark
3d ii	An answer that includes the following: • correct critical value stated / indicated in table (1) • calculated value is less than the critical value, therefore accept the null hypothesis (1) • there is no (significant) difference between the number of mayfly nymphs in streams A and B (1)	2.05	Exp 3

Question Number	Answer	Additional Guidance	Mark
3e	An explanation that includes two of the following: - comment on the variability of data (1) {samples should have been taken on more than one day / at more than one time of day / should have surveyed more than 2 streams} (1) - other named variables not monitored / measured (1)	Accept range bars overlap Accept number of species of mayfly nymphs should have been recorded Accept depth / flow rate / temperature / pH / light intensity / other pollutants	Exp 2

(Total for question 3 = 15 marks)